

SHADOWFOX

DATA ANALYST INTERNSHIP

FINAL INTERNSHIP REPORT

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Internship Track: Beginner → Intermediate → Advanced

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1. Introduction

This report documents the work completed during the ShadowFox Data Analyst Internship, structured across three progressive levels — Beginner, Intermediate, and Advanced. The internship was designed to evaluate not only technical execution but also data understanding, business interpretation, analytical thinking, reporting clarity, and readiness to work on practical, decision-oriented analysis.

Each level built on the one before it. The Beginner project focused on cleaning and organizing a retail sales dataset in Microsoft Excel and presenting it through a simple spreadsheet dashboard. The Intermediate project moved into deeper customer, revenue, and profitability analysis on a larger e-commerce dataset, combining Excel preparation with an interactive Power BI report. The Advanced project applied the same analytical discipline to an HR dataset, producing an executive-style Power BI dashboard built around KPI cards, slicers, and decision-oriented insights.

Taken together, the three projects trace a single analytical journey: from descriptive spreadsheet reporting, to segment-level business analysis, to interactive executive dashboarding. This report presents that journey as one continuous body of work rather than three unrelated submissions.

2. Internship Objectives

The objectives of the internship, as reflected across the three levels of work, were to develop and demonstrate the following capabilities:

- Understanding and working with real-world structured datasets
- Cleaning, validating, and preparing data for analysis
- Performing exploratory and descriptive analysis
- Selecting meaningful KPIs appropriate to the business question
- Identifying trends and patterns across time, category, and segment
- Communicating findings clearly through charts and dashboards
- Turning analytical findings into practical, evidence-based recommendations
- Progressing from static spreadsheet reporting to interactive Power BI dashboarding
- Developing a progressively stronger analytical and business-oriented mindset

3. Overall Internship Workflow

Across all three projects, the same underlying analytical workflow was followed, though it grew more sophisticated at each level:

- Dataset selection
- Data cleaning and preparation
- Exploratory analysis
- Framing business questions
- KPI development

- Visualization
- Business insight generation
- Recommendation development
- Dashboard / report construction
- Decision support

At the Beginner level, this workflow produced a static Excel dashboard answering "what happened?". At the Intermediate level, the same workflow was applied to a richer dataset and extended into Power BI, answering "what happened, how do segments differ, and what patterns might explain it?". At the Advanced level, the workflow culminated in an interactive, filterable executive dashboard answering "where is risk concentrated, and what should management do about it?"

4. Beginner Level Project — Retail Sales Analysis Dashboard

4.1 Project Overview

The Beginner-level project was completed using Microsoft Excel and focused on analyzing the Madhav E-Commerce Sales Dataset. The goal was to take raw order and transaction data and convert it into a clean, analysis-ready dataset, then use standard spreadsheet techniques to understand overall sales performance, profitability, monthly movement, category and sub-category performance, regional sales, and payment-mode behavior. The final output was a single-sheet Excel dashboard combining KPI cards, charts, business insights, and recommended actions.

4.2 Dataset Used

The dataset was supplied as two related files, joined on a common Order ID field:

Orders (500 records)	Details (1,500 records)
Order ID, Order Date, Customer Name, State, City	Order ID, Amount, Profit, Quantity, Category, Sub-Category, Payment Mode

4.3 Data Integration

The Orders and Details files were merged using Order ID into a single working table, Sales_Data. Because Details held 1,500 records against 500 orders, the merged table also contains 1,500 rows — one row per product/detail line within an order. Repeated Order IDs in the merged table are therefore expected and valid, since a single order can include more than one product line; they were not treated as duplicate errors.

4.4 Data Cleaning and Preparation

Before analysis, the merged data was validated and standardized:

- Data types reviewed and corrected across all fields (Order ID kept as text, Order Date formatted as a date, Amount/Profit as numeric, Quantity as a whole number)
- Missing-value checks performed across the dataset
- Genuine duplicate rows checked and treated separately from the expected repeated Order IDs
- Column names cleaned for readability (e.g. CustomerName → Customer Name)
- The cleaned table (SalesData) organized as a structured Excel table to support the analysis

4.5 Analysis Performed

Using the cleaned dataset, the following analyses were carried out on a dedicated Analysis sheet:

- Core summary metrics: Total Sales, Total Profit, Total Quantity, Total Orders (based on unique Order IDs, not row count), and Average Order Value
- Monthly sales, profit, and quantity trends
- Category-level performance (sales, profit, quantity)
- Sub-category-level performance

- State-level sales and profit
- Payment-mode analysis by sales and quantity

4.6 Dashboard

The final Dashboard sheet brought together KPI cards (Total Sales, Total Profit, Total Orders, Average Order Value), a Monthly Sales Trend chart, Sales by Category, Sales by Payment Mode, and Sales by Sub-Category charts, along with a written Business Insights section and a Recommended Actions section. The dashboard was kept deliberately simple and uncluttered, in keeping with the Beginner-level scope of the task.

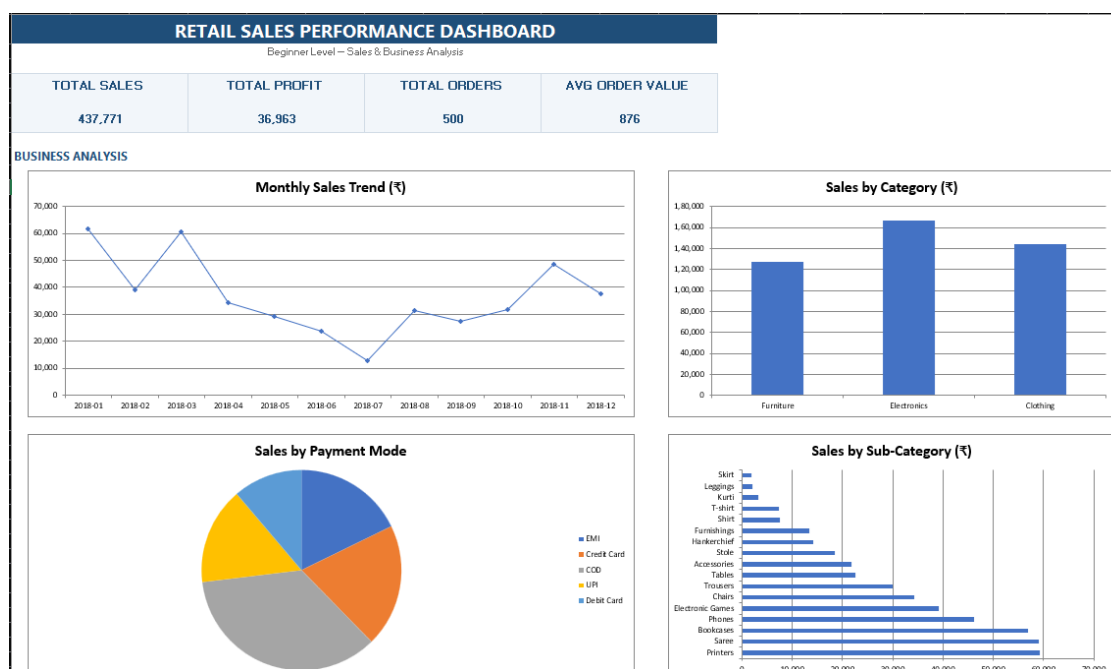


Figure 1. Retail Sales Analysis Dashboard developed in Microsoft Excel

4.7 Beginner Findings

Finding	Value
Total Sales	₹437,771
Total Profit	₹36,963
Total Orders	500
Average Order Value	₹876
Strongest month	January 2018 — approx. ₹61,632
Weakest month	July 2018 — approx. ₹12,966
Highest-sales category	Electronics — approx. ₹166,267
Highest-profit category	Clothing — approx. ₹13,325
Strongest sub-category	Printers — approx. ₹59,252 in sales
Leading state	Maharashtra — approx. ₹102,498

Finding	Value
Most-used payment mode (by quantity)	COD — approx. 2,456 units

4.8 Beginner Recommendations

- Plan promotions and inventory around the stronger sales periods observed in the monthly trend
- Maintain sufficient stock and promotional attention for high-performing categories such as Electronics
- Review strong sub-category demand, especially Printers, for possible cross-selling and bundling
- Factor regional and payment-mode preferences into sales and customer-facing planning

4.9 Beginner Learning Outcome

This level demonstrated the core foundations of spreadsheet-based data analysis: joining related tables on a key field, validating and cleaning data types, correctly handling repeated identifiers, selecting appropriate summary metrics, choosing suitable chart types, and structuring a simple dashboard that a non-technical reader could interpret at a glance.

5. Intermediate Level Project — Global E-Commerce Customer & Revenue Analytics

5.1 Project Overview

The Intermediate-level project was intentionally more analytical than the Beginner project, moving from simple summary reporting into customer behavior, segment profitability, discount impact, and regional performance. It was built using Excel for data preparation and Microsoft Power BI for the final interactive report, working with a Global E-Commerce Sales & Customer dataset.

5.2 Dataset Profile

Metric	Value
Rows	2,000
Original columns	15
Missing values	0
Fully duplicated rows	0
Countries	20
Regions	5
Product categories	4
Products	40
Unique customers	1,534
Unique orders	2,000

5.3 Data Fields

The dataset included Order_ID, Order_Date, Customer_Name, Customer_Segment, Country, Region, Product_Category, Product_Name, Quantity, Unit_Price, Discount_Percent, Total_Sales, Shipping_Cost, Profit, and Payment_Method.

5.4 Data Cleaning and Preparation

Preparation in Excel included data quality profiling, data-type validation, text standardization, blank-value checks, and full-row duplicate checks. Several derived fields were created to support deeper analysis:

- Order Year and Order Month / Order Month Name — for time-based trend analysis
- Profit Margin % — Profit divided by Total_Sales, for profitability analysis
- Discount Band — grouping Discount_Percent into ranges for margin-impact analysis

5.5 Exploratory Analysis — Key Metrics

KPI	Value
Total Sales	\$484,559.34
Total Profit	\$158,872.32
Total Quantity	7,115
Total Orders	2,000
Unique Customers	1,534
Average Order Value	\$242.28
Average Profit per Order	\$79.44
Average Discount	approx. 8.57%

5.6 Deep Analysis

Customer Behavior

- Repeat customers: 392
- One-time customers: 1,142
- Repeat customer rate: 25.55%
- Top 10 customers' share of total revenue: approx. 6.35%

Customer Segment Profitability

Segment	Sales	Profit	Margin
Consumer	≈ \$256.29K	\$87.30K	≈ 34.06%
Corporate	≈ \$146.05K	\$44.46K	≈ 30.44%
Home Office	≈ \$82.22K	\$27.11K	≈ 32.97%

Discount Impact on Margin

Discount Band	Profit Margin
No Discount	38.72%
0–5%	35.36%
6–10%	31.93%
11–15%	27.02%
16–20%	21.39%

Margins declined steadily as discount depth increased. This is reported as an association observed in the data, not a proven causal relationship.

Regional Performance

- Europe: approx. \$137.01K in sales
- North America: approx. \$133.88K in sales

- North America recorded the strongest regional margin, at approx. 33.80%
- Europe's margin was approx. 33.34%

5.7 Power BI Report

The Power BI report was organized into three pages, moving from a high-level executive view down to segment- and product-level detail:

Page 1 — Global E-Commerce | Executive Overview

Revenue, profit, and margin KPI cards; customer count and repeat rate; monthly sales trend; sales by customer segment and by region; and slicers for interactive filtering.

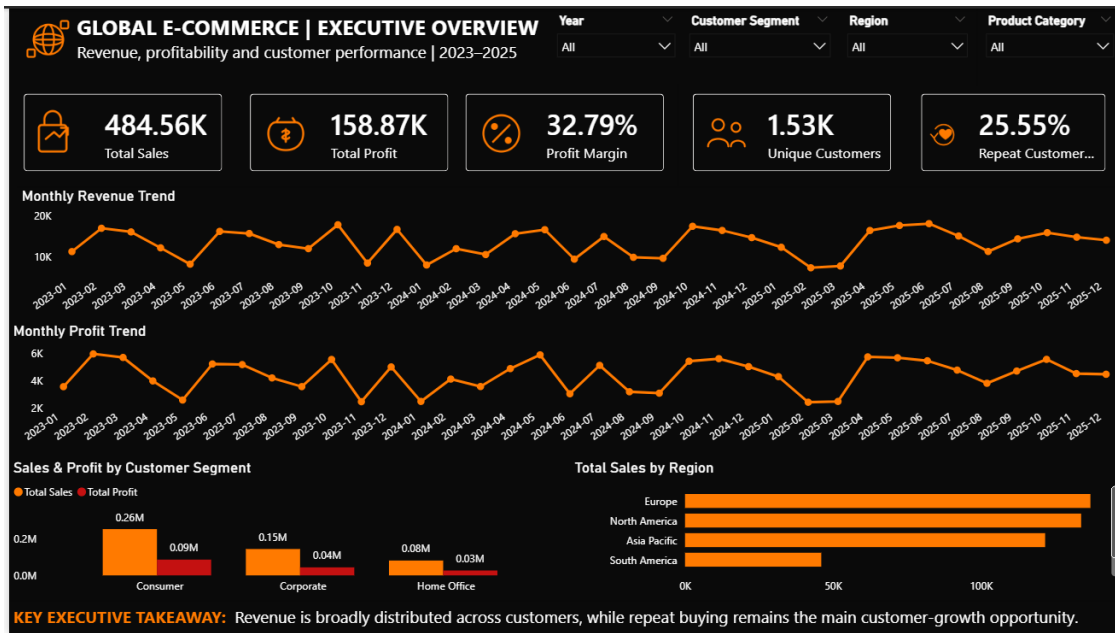


Figure 3. Global E-Commerce | Executive Overview — Power BI report page 1

Page 2 — Customer & Segment Analysis

Unique customers, repeat vs. one-time customers, repeat customer rate, sales and profit by segment, profit margin by segment, and the top 10 customers by revenue.

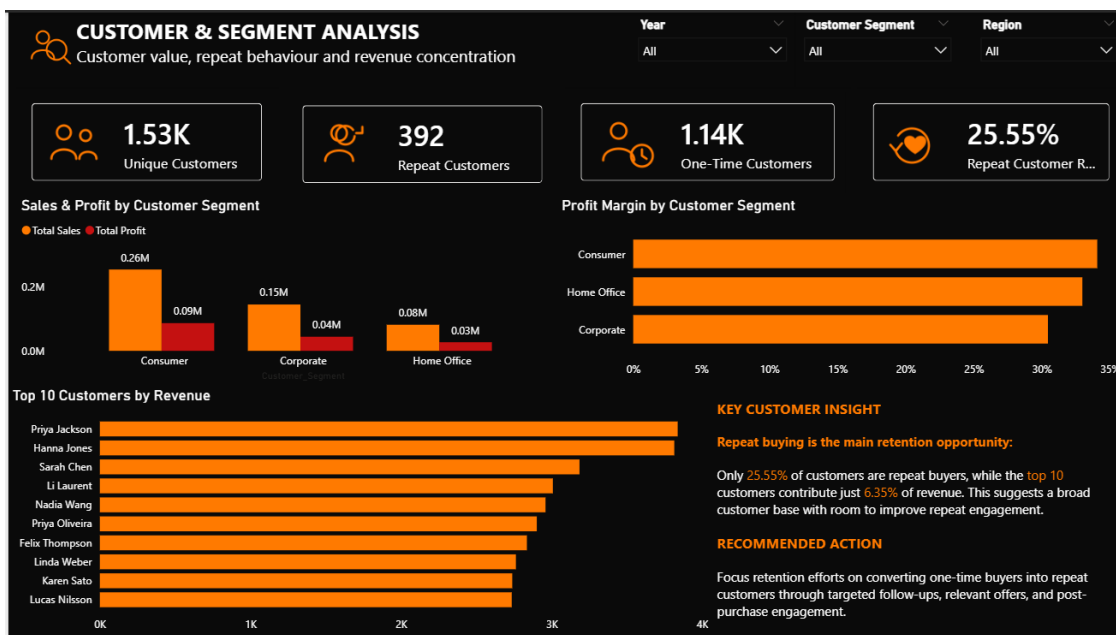


Figure 4. Customer & Segment Analysis — Power BI report page 2

Page 3 — Product & Profitability Analysis

Top products by revenue, sales versus profit margin by product, the impact of discounting on profit margin, and regional revenue and profitability, supported by KPI cards and slicers.

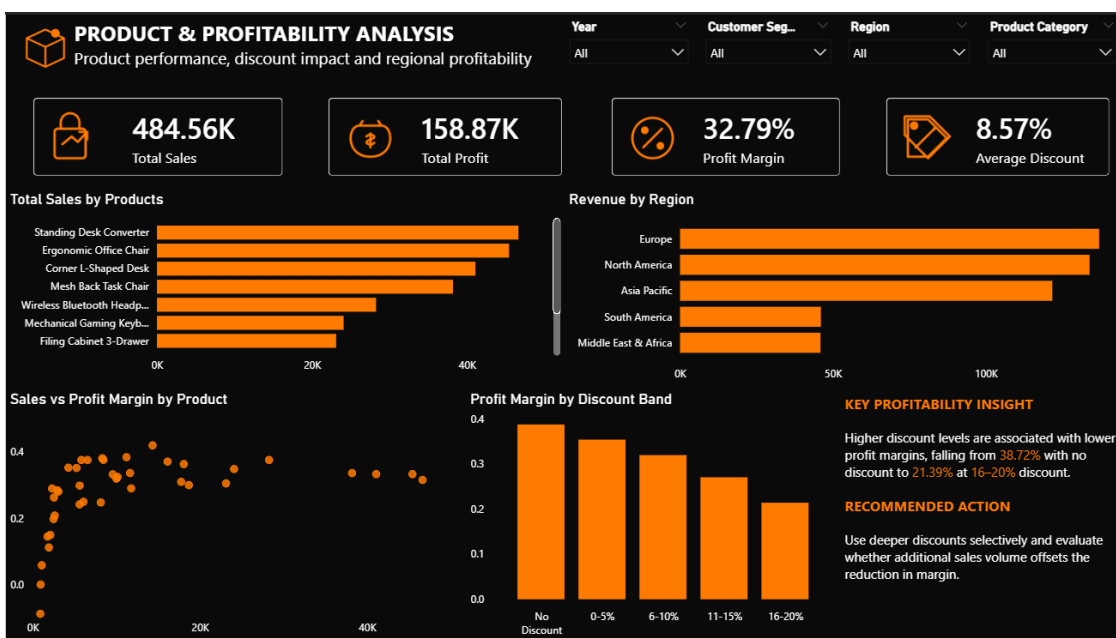


Figure 5. Product & Profitability Analysis — Power BI report page 3

5.8 Intermediate Insights

- Higher discount levels were associated with lower profit margins, most noticeably beyond the 10% discount band

- Roughly one in four customers made a repeat purchase, while the large majority (1,142 of 1,534) purchased only once
- Revenue was not heavily concentrated in a small number of customers — the top 10 customers accounted for only about 6.35% of total revenue
- The Consumer segment generated the most sales and the highest profit margin among the three segments
- North America and Europe were the strongest regions by both sales and profitability

5.9 Intermediate Recommendations

- Strengthen engagement with repeat customers, who show materially higher long-term value
- Target one-time customers with follow-up offers to encourage a second purchase
- Review heavy discounting, particularly bands above 10%, where margin erosion is most pronounced
- Prioritize products and segments that combine strong sales with healthy margins rather than sales volume alone
- Evaluate regional performance using both revenue scale and profitability, since the two do not always align

5.10 Intermediate Learning Outcome

This project marked the shift from descriptive Excel reporting to segment-level business analysis, and from a static spreadsheet dashboard to an interactive Power BI report. It required distinguishing revenue from profitability, interpreting discount and margin patterns carefully (association rather than causation), and building a multi-page Power BI model with reusable measures and slicers.

6. Advanced Level Project — Employee Attrition & Workforce Analytics

6.1 Project Overview

The Advanced-level project was built end-to-end in Microsoft Power BI, using its Report View, Data/Model View, DAX measures, interactive slicers, and dashboard formatting tools. It applied an executive-dashboard design approach to an HR analytics problem: turning raw employee-level records into an interactive tool that HR stakeholders could use to see where attrition risk was concentrated.

6.2 Dataset

The dashboard is built on a single table, HR_Data, containing employee-level records including Age, Gender, Department, Job Role, Attrition (Yes/No), Monthly Income, OverTime, Years at Company, Job Satisfaction, and other employee attributes.

6.3 Data Preparation

Column creation was kept deliberately minimal, limited to calculated columns that genuinely aided interpretation:

- Age Group — five bands (Under 25, 25–34, 35–44, 45–54, 55+), with a sort-helper column to preserve correct ordering
- Income Group — five bands (Below ₹3K, ₹3K–₹5K, ₹5K–₹7K, ₹7K–₹10K, ₹10K+), with a matching sort column
- Tenure Group — five lifecycle stages (0–1 Year, 2–3 Years, 4–5 Years, 6–10 Years, 10+ Years), with a sort column
- Job Satisfaction Level — numeric scores (1–4) converted to readable labels (1 - Low ... 4 - Very High)

6.4 KPI Selection and DAX Measures

A dedicated Measures table held reusable DAX measures so every calculation would respond automatically to whatever slicer selections were active:

- Total Employees — `COUNTROWS(HR_Data)`
- Employees Left — count of records where Attrition = "Yes"
- Attrition Rate — `Employees Left ÷ Total Employees`, with a safe zero fallback
- Average Monthly Income and Average Years at Company

Because these were built as measures rather than static calculations, placing a field such as Department on a chart caused Power BI to re-evaluate Attrition Rate automatically for every department in view, without needing a separate formula for each category.

6.5 Power BI Dashboard Design

A dark-background, neon-red-accent theme was applied deliberately to reinforce the dashboard's focus on risk and attrition. The page was organized into four zones: a title bar with three slicers at the top, five KPI cards beneath it, a main analytical area with seven attrition-rate visuals (Department, Job Role, Overtime, Monthly

Income, Tenure, Age, Job Satisfaction), and a right-side Key Insights & Recommendations panel — kept in a dedicated panel rather than a second page, so the dashboard itself answers "what is happening" while the panel answers "what does it mean."

Dashboard View

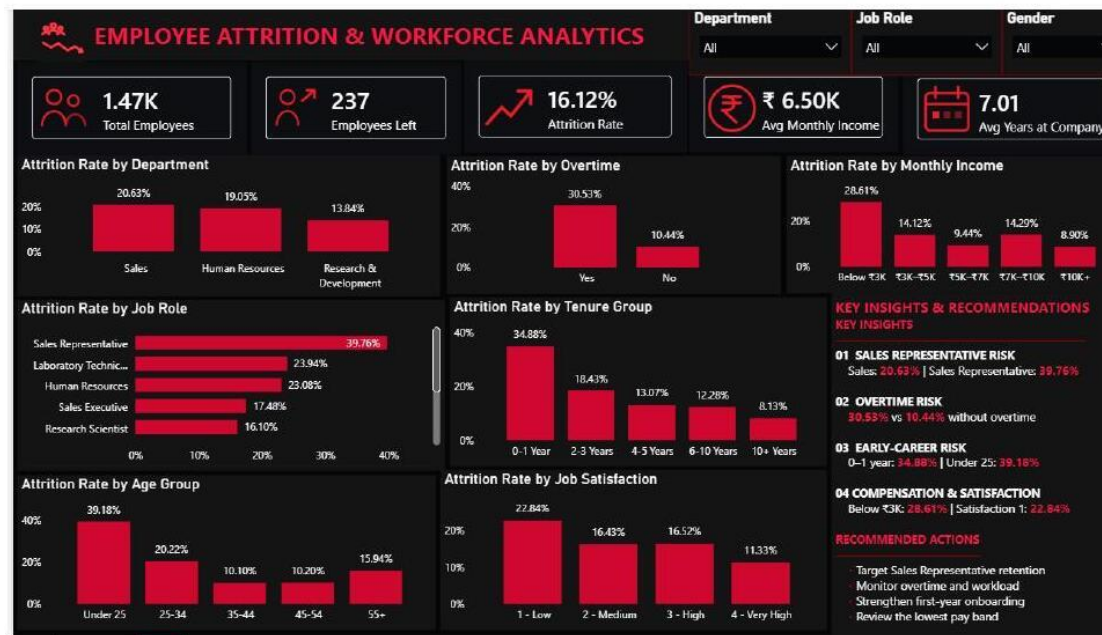


Figure 2. Employee Attrition & Workforce Analytics dashboard — final Power BI build (Dark + Neon Red theme)

6.6 Interactive Filters / Slicers

Three slicers — Department, Job Role, and Gender — let a user drill into any subgroup, with every KPI card and visual on the page updating automatically through Power BI's filter context. Each slicer was individually tested to confirm it correctly updated the relevant KPIs and visuals.

6.7 KPI Analysis

KPI	Value
Total Employees	1.47K
Employees Left	237
Attrition Rate	16.12%
Average Monthly Income	₹6.50K
Average Years at Company	7.01

6.8 Business Insights

- Sales Representatives showed the highest role-level attrition, at 39.76%, against a Sales department-wide rate of 20.63%
- Employees working overtime showed a 30.53% attrition rate, compared with 10.44% for those who did not

- Employees with 0–1 year of tenure showed 34.88% attrition, and employees under 25 showed 39.18%
- Employees earning below ₹3K monthly showed 28.61% attrition
- Employees reporting the lowest job satisfaction level showed 22.84% attrition

As documented in the project's Limitations section, these relationships are reported as associations found in the data — overtime and income are described as being associated with higher attrition, not as proven causes of it, since the analysis is descriptive and diagnostic rather than a causal or predictive study.

6.9 Recommendations

- Target Sales Representative retention specifically — role expectations, incentives, workload, and career progression — rather than a Sales-department-wide intervention
- Monitor overtime patterns and workload distribution, with particular attention where overtime overlaps with other risk factors
- Strengthen first-year onboarding through structured check-ins, mentoring, and early-career development paths
- Review the lowest compensation band specifically, considering targeted retention or compensation action rather than an across-the-board raise

6.10 Advanced Learning Outcome

This project demonstrated the ability to design and build a genuinely interactive, decision-support dashboard rather than a static chart collection: selecting a focused set of KPIs, building reusable DAX measures that respond correctly to filter context, applying a deliberate visual theme, and testing slicers and calculations before considering the work complete.

7. Comparison of the Three Levels

Area	Beginner	Intermediate	Advanced
Project	Retail Sales Analysis	Global E-Commerce Customer & Revenue Analytics	Employee Attrition & Workforce Analytics
Main tool(s)	Microsoft Excel	Excel + Power BI	Power BI
Dataset complexity	Two linked tables, 500/1,500 records	Single table, 2,000 rows, 15 fields	Single table, employee-level HR records
Analysis depth	Descriptive summary metrics and trends	Customer, segment, and profitability analysis	Multi-dimensional attrition-risk analysis
KPI complexity	4 core KPIs	8+ KPIs incl. margin and repeat-rate	5 KPIs driven by reusable DAX measures
Visualization	Static Excel charts	Multi-page Power BI report	Executive-style Power BI dashboard, custom theme
Interactivity	None	Slicers within Power BI pages	Three interactive slicers with full filter-context response
Business interpretation	"What happened?"	"What happened, and why might it be happening?"	"Where is risk concentrated, and what should be decided?"
Recommendations	General business actions	Segment- and discount-based recommendations	Targeted, role- and factor-specific HR actions
Reporting style	Single-sheet dashboard	Structured report with narrative insights	Executive dashboard with dedicated insights panel

8. Skills Developed

The internship progressively built the following skills:

- Data cleaning and preparation (type validation, missing-value and duplicate checks, text standardization)
- Excel-based spreadsheet analysis and dashboarding
- Exploratory and descriptive analysis
- KPI development and selection
- Customer, revenue, and profitability analysis
- Trend analysis across time, category, and segment
- Power BI report and dashboard design
- DAX measures and reusable calculations
- Interactive slicers and filter-context design
- Business storytelling, insight generation, and recommendation writing

These skills developed cumulatively: Excel fundamentals from the Beginner project underpinned the more complex data preparation required at the Intermediate level, and the segment-level analytical thinking developed there carried directly into the KPI and DAX work of the Advanced Power BI dashboard.

9. Challenges and Solutions

Beginner

- Integrating the Orders and Details datasets correctly on Order ID
- Recognizing that repeated Order IDs were valid (multiple products per order) rather than duplicate errors, and calculating Total Orders from unique IDs rather than row count
- Organizing the workbook so raw data, working data, and final dashboard stayed clearly separated

Intermediate

- Preparing a larger, multi-dimensional dataset and deriving useful analytical fields (profit margin, discount band, order year/month)
- Distinguishing revenue performance from profitability performance, since the two did not always align by segment or region
- Interpreting discount-margin patterns as an association rather than overstating them as a causal relationship
- Building a multi-page Power BI model and maintaining a focused visual hierarchy across pages

Advanced

- Selecting a focused set of HR KPIs rather than trying to surface every possible metric
- Building interactive slicers and confirming that every visual correctly responded to filter context
- Balancing visual density with readability on a single-page executive dashboard
- Translating attrition patterns into specific, targeted recommendations rather than generic HR advice

10. Business and Analytical Approach

The internship showed a clear progression in analytical framing across the three levels:

Beginner: "What happened?" — descriptive summary of sales, profit, and trends.

Intermediate: "What happened, how do segments differ, and what patterns may explain the business result?" — customer, segment, and discount-margin analysis.

Advanced: "What is happening, where is risk concentrated, and what decisions should management consider?" — executive-level, decision-oriented HR analytics.

11. Learning Outcomes

Across the internship, the work strengthened practical data-handling skills — correctly joining and cleaning multi-table datasets, validating data types, and organizing raw data into analysis-ready structures. It also

strengthened analytical judgment: choosing the right summary metrics, separating revenue from profitability, and interpreting patterns like discount impact and overtime-related attrition carefully, as associations rather than proven causes.

On the technical side, the internship built confidence in Excel-based spreadsheet analysis and dashboarding, and then extended that into Power BI — building a data model, writing reusable DAX measures, and designing interactive, slicer-driven dashboards. Throughout, the emphasis was on communicating findings clearly: through KPI cards, well-chosen charts, and concise written insights that connect a number to a business action. The three projects did not simply repeat the same skills at different scales — each one required a genuinely deeper level of business interpretation than the one before it.

12. Overall Business Insights

Read together, the three projects show a consistent analytical thread: identifying where performance or risk is concentrated, rather than only reporting totals. The Beginner project identified which categories, sub-categories, and regions drove sales. The Intermediate project went further, showing that revenue leaders and profitability leaders are not always the same thing, that a small minority of customers were not disproportionately responsible for revenue, and that discounting had a measurable, non-causal association with margin erosion. The Advanced project applied the same discipline to a workforce question, isolating specific high-risk segments (a single job role, early tenure, overtime status, and the lowest income band) rather than treating attrition as a single undifferentiated number.

Together, the three projects demonstrate the ability to move from a raw dataset to a segmented, evidence-based understanding of where performance or risk actually sits, and to translate that understanding into recommendations a business or HR team could act on.

13. Overall Recommendations

- Use category- and product-level performance data (Beginner, Intermediate) to guide inventory and promotional planning, rather than relying on top-line sales alone
- Treat discounting as a lever to be managed carefully — the Intermediate project's findings suggest reviewing discount depth beyond 10% specifically
- Prioritize retention efforts by segment rather than department-wide: the Advanced project shows attrition risk concentrated in specific roles, tenure bands, and income groups, not spread evenly across the workforce
- Continue building on interactive, slicer-driven dashboards (as in the Intermediate and Advanced projects) so that stakeholders can explore data themselves rather than requesting a new static report for each question

14. Conclusion

The ShadowFox Data Analyst Internship provided a structured progression from spreadsheet-based analysis, through deeper business and customer analytics, to executive-style Power BI dashboarding. Each level required a real increase in analytical depth and technical capability: the Beginner project established the fundamentals of

data cleaning and spreadsheet reporting; the Intermediate project introduced segment-level business analysis and interactive Power BI reporting; and the Advanced project applied that experience to build a genuinely interactive, decision-support HR dashboard.

Overall, the internship strengthened practical skills in data preparation, analysis, visualization, interpretation, and business communication — the core capabilities expected of a working Data Analyst.

15. Acknowledgment

I would like to thank ShadowFox for the opportunity to complete this Data Analyst Internship and for structuring the program in a way that allowed for genuine progression across three levels of increasing analytical depth. I would also like to thank Amrita Vishwa Vidyapeetham, Amaravati, for the academic foundation that supported this work. This internship was a valuable, practical extension of my classroom learning in data analysis and business interpretation.

16. References / Source Note

- ShadowFox Data Analyst Internship Task List (official program documentation)
- Retail Sales Analysis Dashboard — Beginner Level Project Documentation (Madhav E-Commerce Sales Dataset)
- Global E-Commerce Customer & Revenue Analytics — Intermediate Level project workbook and Power BI file
- Employee Attrition & Workforce Analytics Dashboard — Advanced Level Project Documentation (HR_Data)